

Marine microbiology

Lectures 13 – 14

Introduction to Marine Microbes

References:

Munn. Marine Microbiology (**MM**): Chapt. 1; Chapt. 3, p.79, 84-85.

Prescott's Microbiology (**PM**): Chapt. 7, p.150-151; Chapt. 30,
p.660 -672

Introduction

Outcomes:

At the end of these lectures you will have a broad understanding of the marine environment as a microbial habitat, and be able to describe:

1. Diversity in size of marine microbes
2. Physical and chemical factors important for marine microbes
3. Phytoplankton classification
4. Marine snow
5. Biofilms – structure, development and properties
6. Some important marine environments
 - Sediments (沉积物), sea ice, hydrothermal vents (深海热泉)

Introduction

- Progress in understanding the diversity of microbial life in the sea was slow until relatively recent times
- Marine microbiology is now one of the most rapidly-moving areas of microbiological research, due to:
 - Powerful new tools in molecular microbiology
 - Remote sensing technology
 - Advances in deep-sea exploration
- Now realised that marine microbes play a vital role in the maintenance of the planet – significance includes:
 - Climate change
 - Ocean acidification
 - Marine pollution
 - Marine biotechnology
 - Aquaculture (水产养殖) and disease
 - Marine food webs



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Diversity

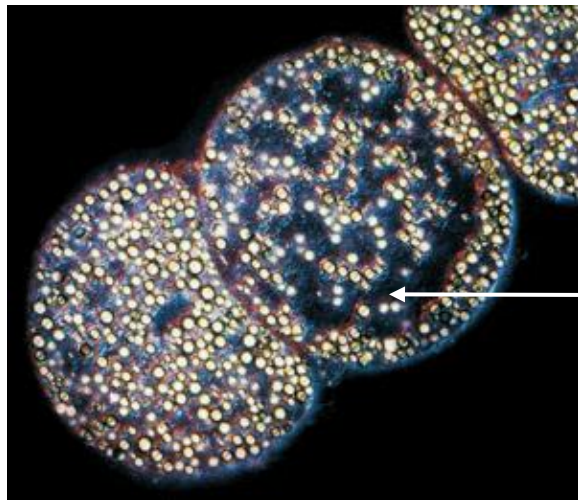
- Marine microbes are found in all three domains of life, and also include viruses
- They show great variation in size

Organism	Characteristics	Size (μm)	Volume (μm ³)
Parvovirus	Icosahedral DNA virus infecting shrimp	0.02	0.000004
Coccolithovirus	Icosahedral DNA virus infecting <i>Emiliana huxleyi</i>	0.17	0.003
<i>Thermodiscus</i>	Disc-shaped hyperthermophilic archaeon	0.2 x 0.08	0.003
<i>Pelagibacter ubique</i>	Crescent-shaped bacterium in plankton	0.1 x 0.9	0.01
<i>Prochlorococcus</i>	Cocci. Dominant photosynthetic ocean bacterium	0.6	0.1
<i>Ostreococcus</i>	Cocci. Prasinophyte alga . Smallest eukaryote	0.8	0.3
<i>Vibrio</i>	Curved rods. Coastal bacterium , animal-associated	1 x 2	2
<i>Pelagomonas calceolata</i>	Photosynthetic flagellate adapted to low light	2	24
<i>Pseudo-nitzschia</i>	Pennate diatom , produces toxin	5 x 80	1600
<i>Staphylothermus marinus</i>	Cocci. Hyperthermophilic archaeon	15	1800
<i>Lingulodinium polyedrum</i>	Bioluminescent bloom-forming dinoflagellate	50	65000
<i>Epulopiscium fishelsoni</i>	Rods. Bacterium symbiotic in fish gut	80 x 600	3000000
<i>Thiomargarita namibiensis</i>	Cocci. Sulfur bacterium	300	14137100

MM from Table 1.1

Diversity: size

- The most abundant marine microbes are *exceptionally* (异常地) small
 - This is the main reason why their abundance was not known until very recently
 - Small cells feeding by osmotrophy (渗透营养) (= absorption of nutrients) can take up these nutrients by diffusion more efficiently than large cells
 - As nutrients are scarce (缺乏的) in oceans, selection has favoured small cells
 - But a few marine bacteria are bigger than many protists e.g. *Thiomargarita namibiensis* 纳米比亚嗜硫珠菌



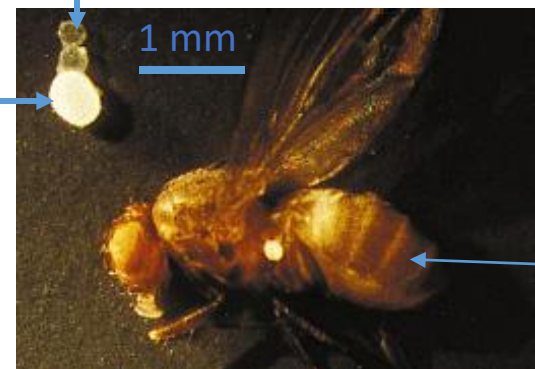
microbewiki.kenyon.edu

Thiomargarita namibiensis

Sulfur globules

Thiomargarita namibiensis visible to the naked eye

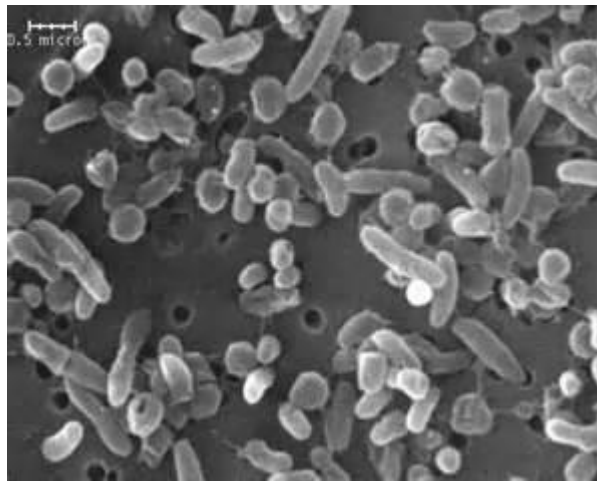
Dead cells



archives.microbeworld.org

Diversity: size

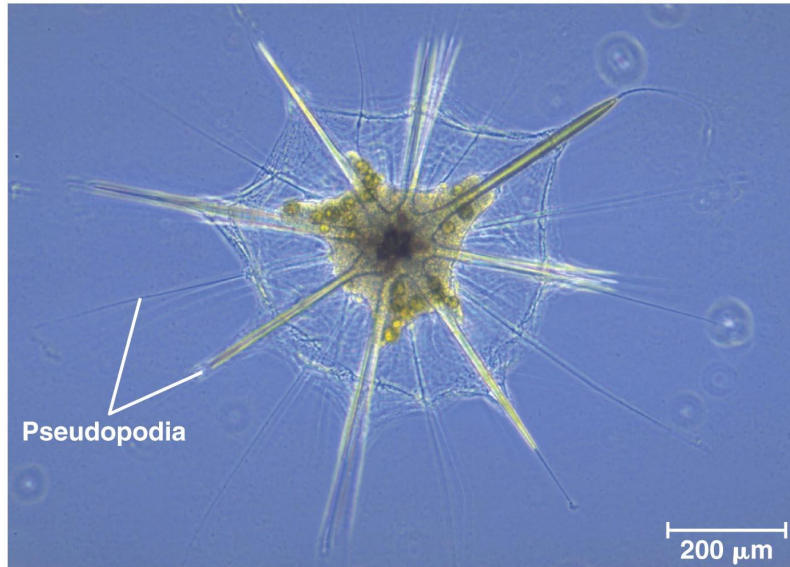
- The small size of “ultramicrobacteria” has been very controversial（有争议的）
 - When marine bacteria are cultured in the lab, starved（饥饿）of nutrients, they become smaller and spherical（球形的）**<0.3 μm** diameter, not metabolically active
 - Is small size of bacteria in oceans due to physiological changes because of starvation? Are they really normal-sized bacteria?
 - Studies showed that addition of nutrients does not cause an increase in size, and small marine bacteria have high metabolic activity
 - Now believed that small size is a genotypically determined property of marine bacteria. They evolved to be small, to ensure efficient use of scarce nutrients in oceans.



Ultramicrobacteria can easily pass through these holes in a membrane filter, while most bacteria cannot

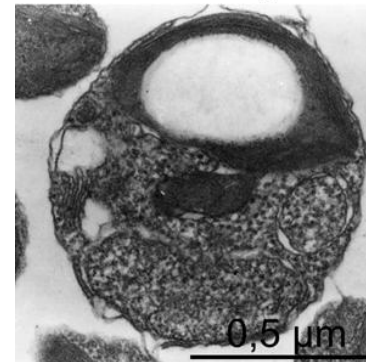
Diversity: size

- Marine eukaryotes are variable in size
 - Many flagellates are 1-2 μm , more typical of bacteria
 - Smallest known eukaryote *Ostreococcus tauri*(一种微藻) is a marine alga
 - These small eukaryotes now called picoeukaryotes(超微型真核生物)
 - 0.2 – 2.0 μm
 - Some members of amoeboid(变形虫)groups (radiolarians放射虫, foraminifera有孔虫) can be visible to the naked eye, but mainly 0.1 – 0.3 mm



A radiolarian

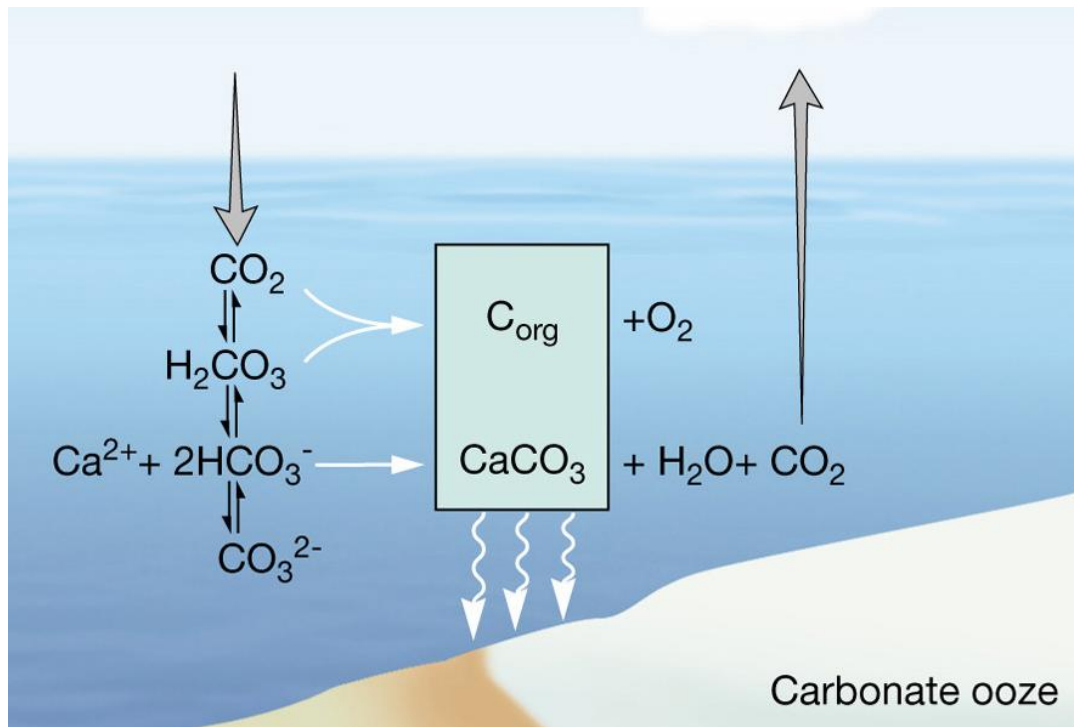
Ostreococcus tauri
"pico"eukaryote



Seawater as a microbial habitat: depends on a number of physical and chemical factors

1. pH and Carbon

- Seawater is slightly **alkaline** (pH 7.5-8.4)
- It is **buffered**(缓冲) by the balance of CO_2 , bicarbonate HCO_3^- , carbonate CO_3^{2-}



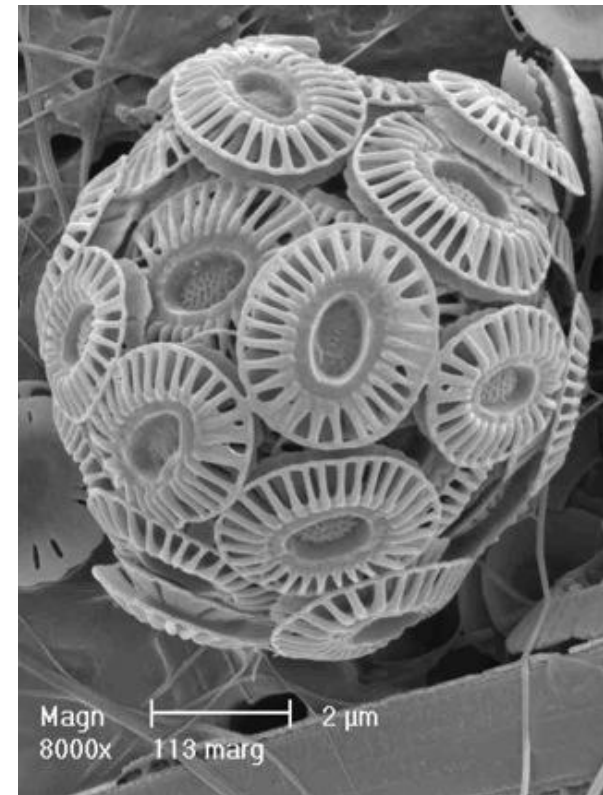
PM Fig. 30.1

- Atmospheric CO_2 enters the oceans
- either converted to organic C by photosynthesis
- or reacts with seawater to form carbonic acid H_2CO_3 which quickly forms bicarbonate (碳酸氢盐) and carbonate (碳酸盐)
- Insoluble (不溶的) CaCO_3 precipitates (沉淀) to form carbonate ooze (软泥)

Seawater as a microbial habitat: depends on a number of physical and chemical factors

1. pH and Carbon

- Increases in atmospheric CO₂ have led to **ocean acidification**, due to the increase in H⁺ ions
 - The average pH of the ocean has dropped by 0.1 since the pre-industrial age
 - pH could fall to 7.9 by 2100
 - One main effect is on reduced formation of shells and skeletons by organisms that use CaCO₃
 - Animals e.g. corals(珊瑚), crustaceans(甲壳类动物)
 - Microbes e.g. coccolithophores(球石藻) (protists)
- Emiliana huxleyi*, an abundant marine coccolithophore



Seawater as a microbial habitat: depends on a number of physical and chemical factors

2. Light and temperature

A. Light is of fundamental importance to photosynthetic microbes

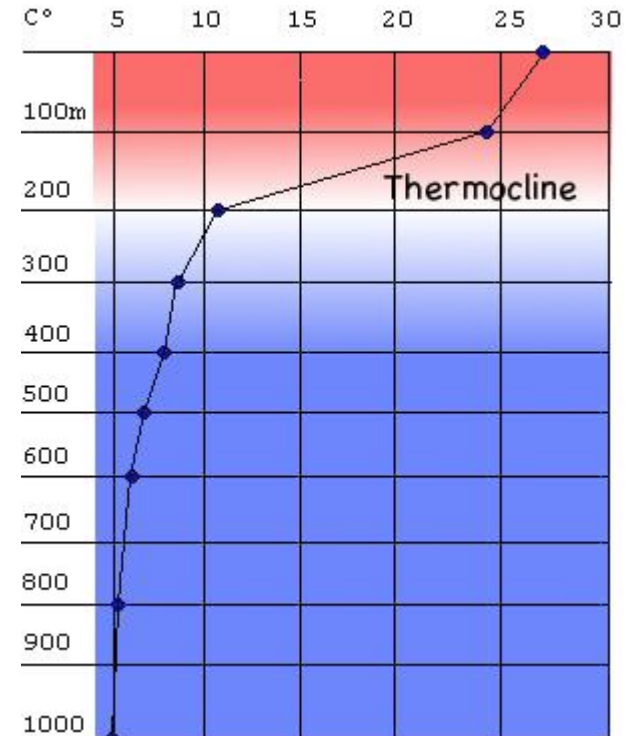
- The depth to which light penetrates (穿透) seawater depends on
 - Cloud cover
 - Polar ice caps (极地冰川)
 - Dust in atmosphere
 - Season and location
- At most, light is sufficient for photosynthesis in **upper 150-200m**:
 - **Photic zone (透光层)** = upper layer of ocean water penetrated by light
 - **Euphotic zone (真光层)** = zone with sufficient light of wavelengths suitable for photosynthesis

Seawater as a microbial habitat: depends on a number of physical and chemical factors

2. Light and temperature

- A. Light is of fundamental importance to photosynthetic microbes
- B. Solar radiation warms the water, and can lead to **thermal stratification** (热分层现象)

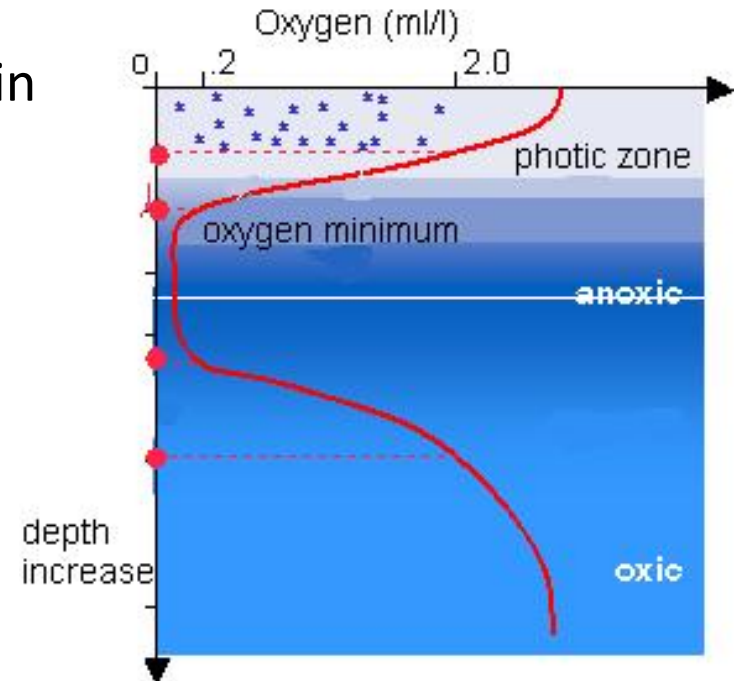
- Warm water is less dense than cool water
- Thermocline (温跃层) develops: mass of warmer water “floating” on top of cooler water
- Layers can mix due to severe storms, or colder temperatures cooling the upper layer (Autumn in temperate climates)
- With mixing, nutrients can move from lower, darker waters to surface – trigger sudden increase in microbes e.g. algal bloom
- Temperature and mixing can have a big effect on microbial activities



Seawater as a microbial habitat

3. Oxygen concentration

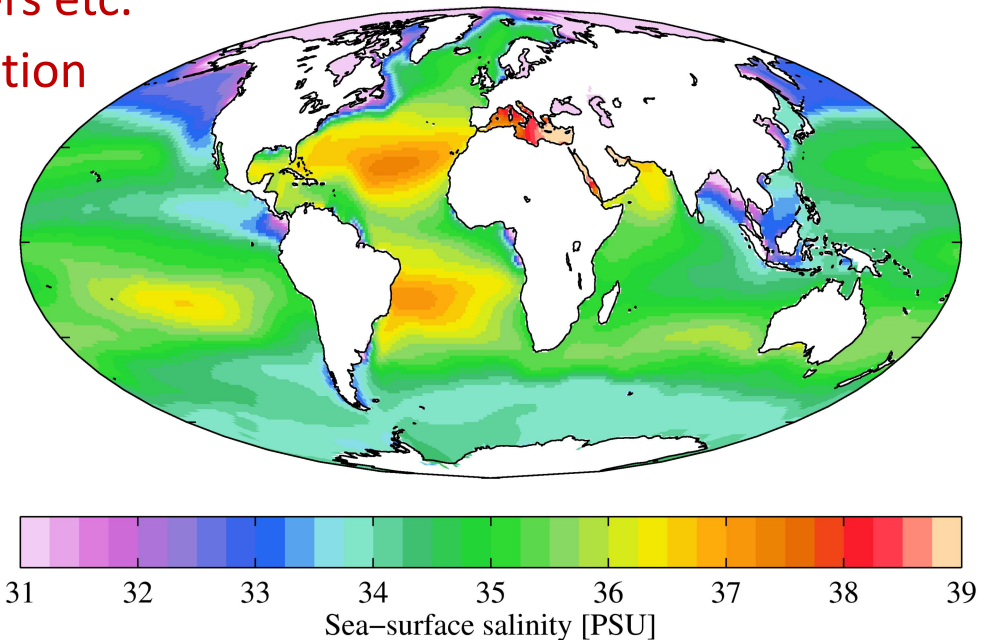
- Oxygen is at its highest concentrations in the top 1-20 m of water
 - exchange with atmosphere
 - production of oxygen by photosynthetic organisms
- Concentration decreases with distance from surface, reaches minimum 200-1000m
 - bacterial decomposition (分解) of organic material may create almost anoxic (缺氧的) conditions
- Below this, oxygen content rises again
 - due to presence of cooler water from polar regions – increased oxygen solubility – see Munn Fig. 1.7 for “global ocean conveyor belt”
 - and increased oxygen solubility at increased pressures in deep ocean



Seawater as a microbial habitat

4. Salinity(盐分)

- Seawater is a complex mixture of >80 solid elements, gases, and dissolved organic substances
- Salinity (parts per thousand ‰) indicates concentration of dissolved substances
 - Open ocean 34-37 ‰, varies with temperature/evaporation(蒸发)/rainfall
 - Coastal 10-32 ‰, dilution by rivers etc.
 - Polar – higher salinity - ice formation
 - Enclosed areas – higher salinity
- Major ionic components 99%
 - Na^+ , Cl^- , SO_4^{2-} , Mg^{2+} , Ca^{2+} , K^+
- Minor ions 1%
 - HCO_3^- , Br^- , borate BO_3^{3-} , Sr^{2+}



Marine microbes form a major component of the plankton

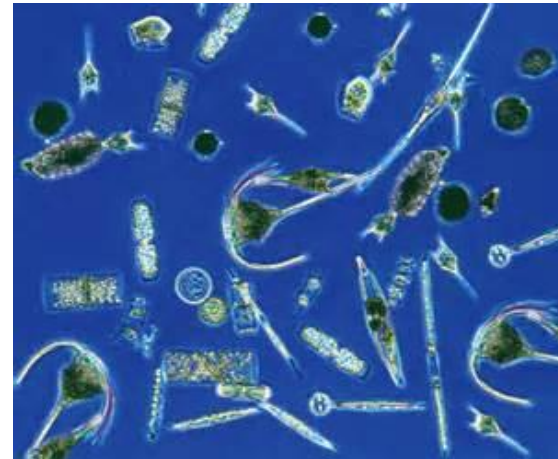
Plankton: general name for free-floating microscopic organisms in water

Definitions traditionally were:

- **Zooplankton:** animals
- **Phytoplankton:** plants

Now the definition is:

- **Phytoplankton:** all free-floating photosynthetic organisms, including diatoms, dinoflagellates and other protists, and cyanobacteria
- Plankton also includes viruses (virioplankton)



Plankton can be classified by size

	Size category	Size range (um)	Microbial groups	
SIZE 	Femtoplankton	0.01-0.2	Viruses	ABUNDANCE 
	Picoplankton	0.2-2	Bacteria* , archaea, some flagellates	
	Nanoplankton	2-20	Flagellates, diatoms, dinoflagellates	
	Microplankton	20-200	Ciliates, diatoms, dinoflagellates, other algae	

*Some filamentous bacteria can occur in larger size classes

MM Table 1.2

Marine snow

- Clumps of material that sink out of the photic zone of the ocean
 - mainly organic
 - the drifting particles resemble snow

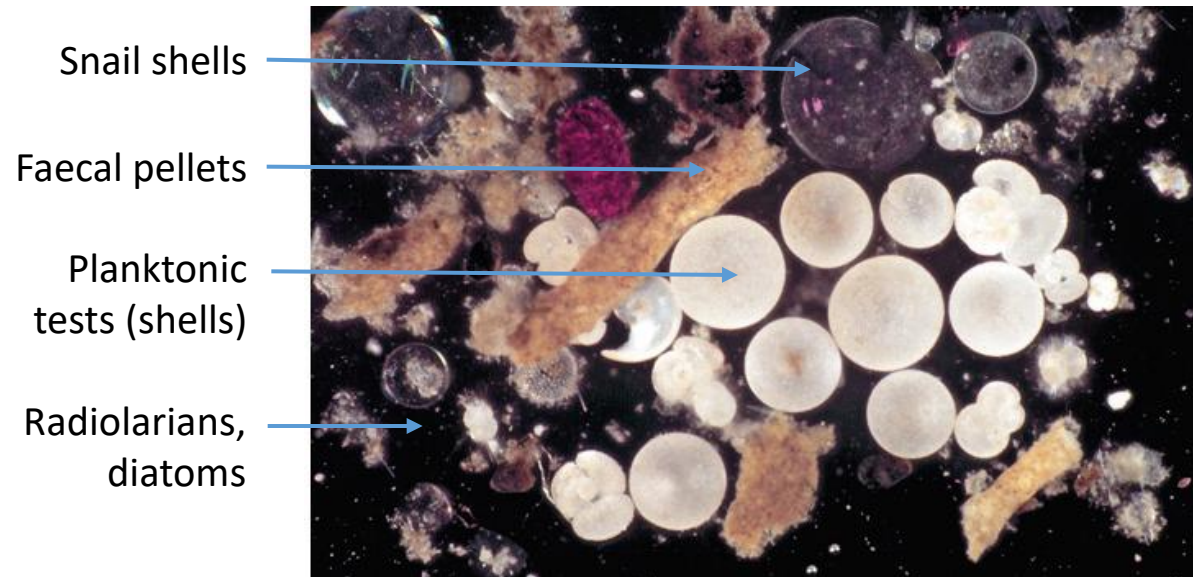


Marine snow

- Particles composed of aggregates of:
 - zooplankton faecal (排泄物) pellets (颗粒),
 - diatom frustules 细胞膜 (cell wall made of silica) – inorganic,
 - detritus 碎屑 (waste or debris) from dead plankton,
 - living plankton cells,
 - glued together by polysaccharide polymers released from phytoplankton, and mucus polysaccharide of zooplankton

**Material collected at 5367 m
at bottom of Sargasso Sea.**

PM Fig. 30.8



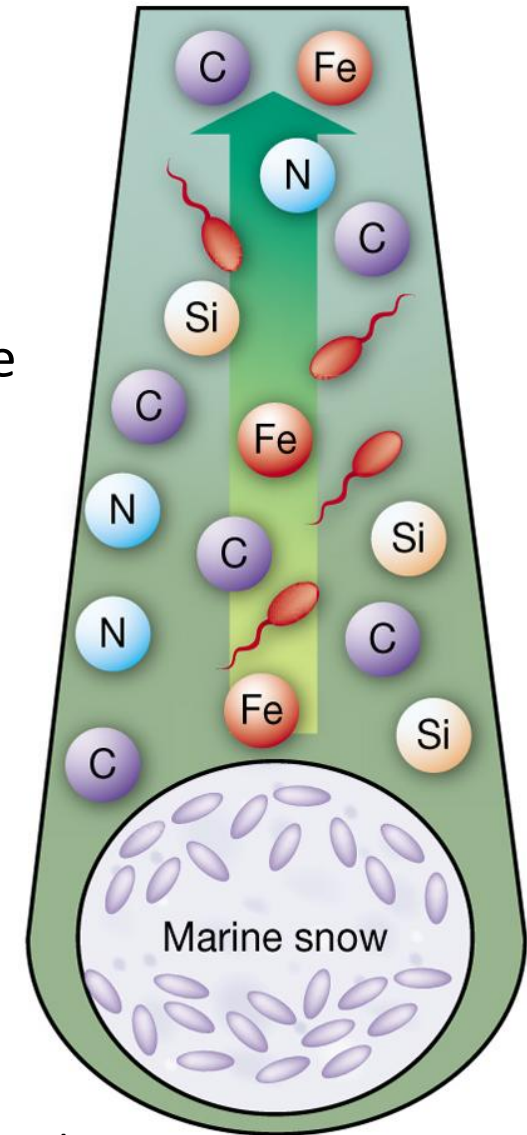
Marine snow

- Particles 0.5-5 μm but some several cm
- Aggregates increase in size as they collide (碰撞)
- Mainly produced in upper 100-200 m
- Sink up to 100 m per day
- During its fall to the bottom of the ocean, marine snow is colonised by microbes
- Microbial populations associated with marine snow can reach **$10^8 - 10^9$ per ml**, much higher (100x to 10000x) than in surrounding water



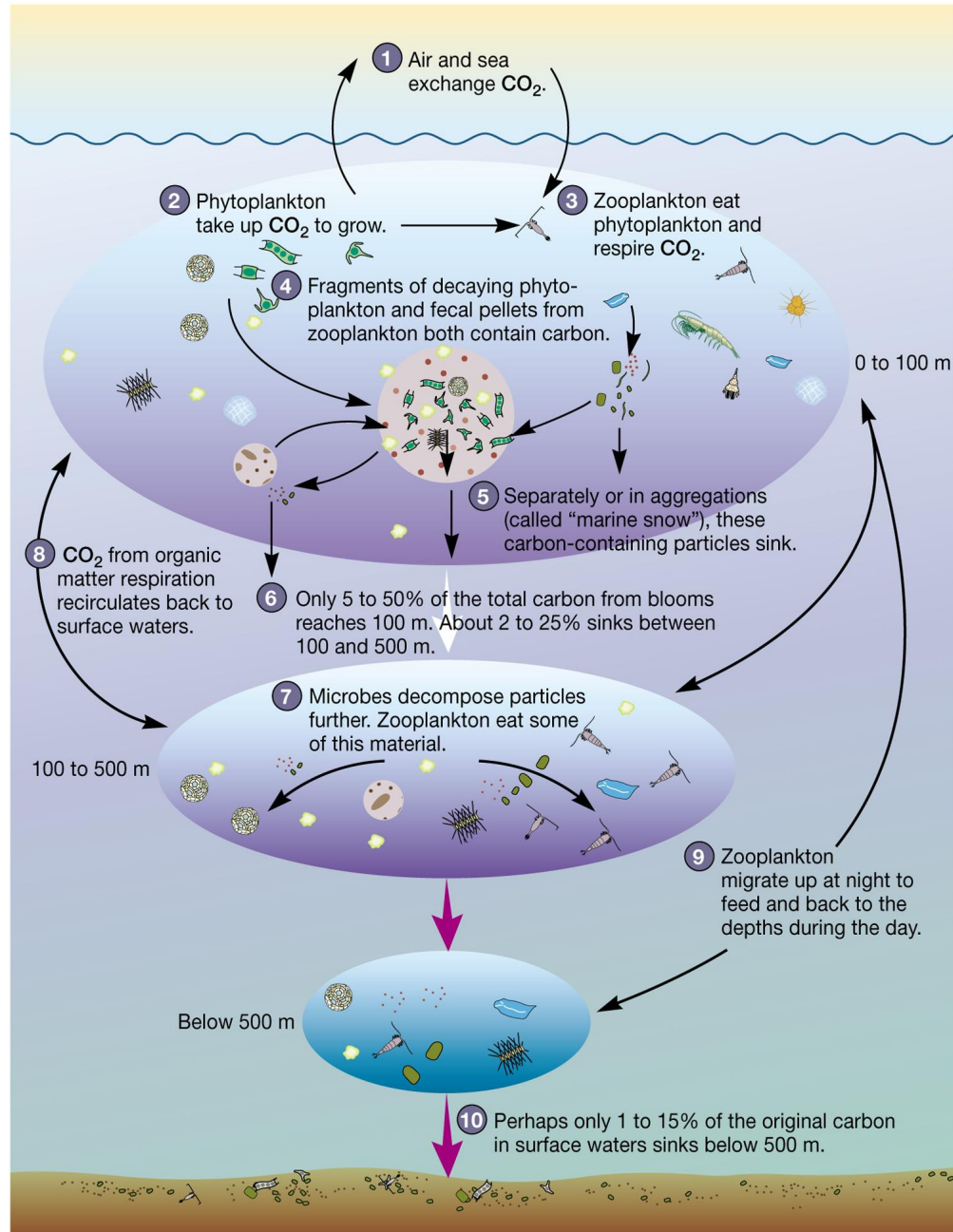
Marine snow

- The microbes secrete (分泌) a variety of exoenzymes 胞外酶 (extracellular enzymes): Exoenzymes break down organic polymers
- Also, viruses lyse (break open) cells to release organic material
- As the marine snow falls, it leaves a **plume** of Dissolved Organic Material (DOM), N, P, Fe
- These nutrients attract chemotactic free-swimming microbes which consume them
- Zooplankton also consume particles
- By the time a piece of snow reaches the seafloor, <1% of original organic C remains



Plume: resembles shape of a feather

The fate of carbon from the photic zone to the benthos (seafloor)



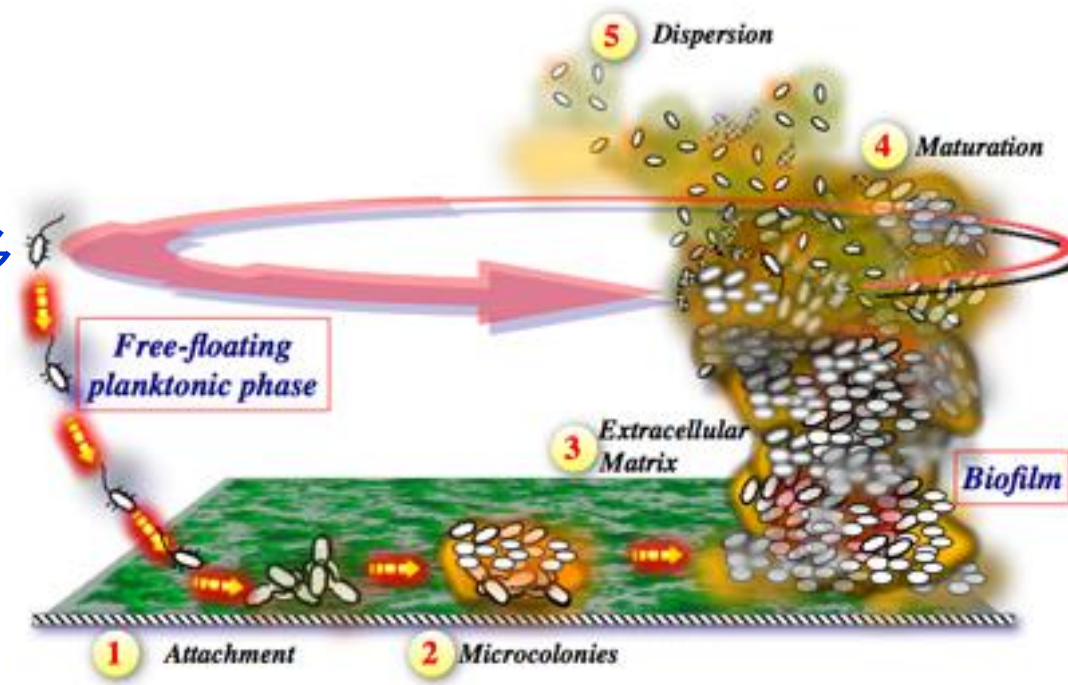
PM Fig. 30.9

Biofilms

- It was only realised during the last 20 years that more microbes live attached to surfaces (sessile) than free-floating (planktonic)
- Microbes often live in “high-rise(高楼)apartments” called biofilms

**A biofilm is a structured/
organised community of
microbial cells**

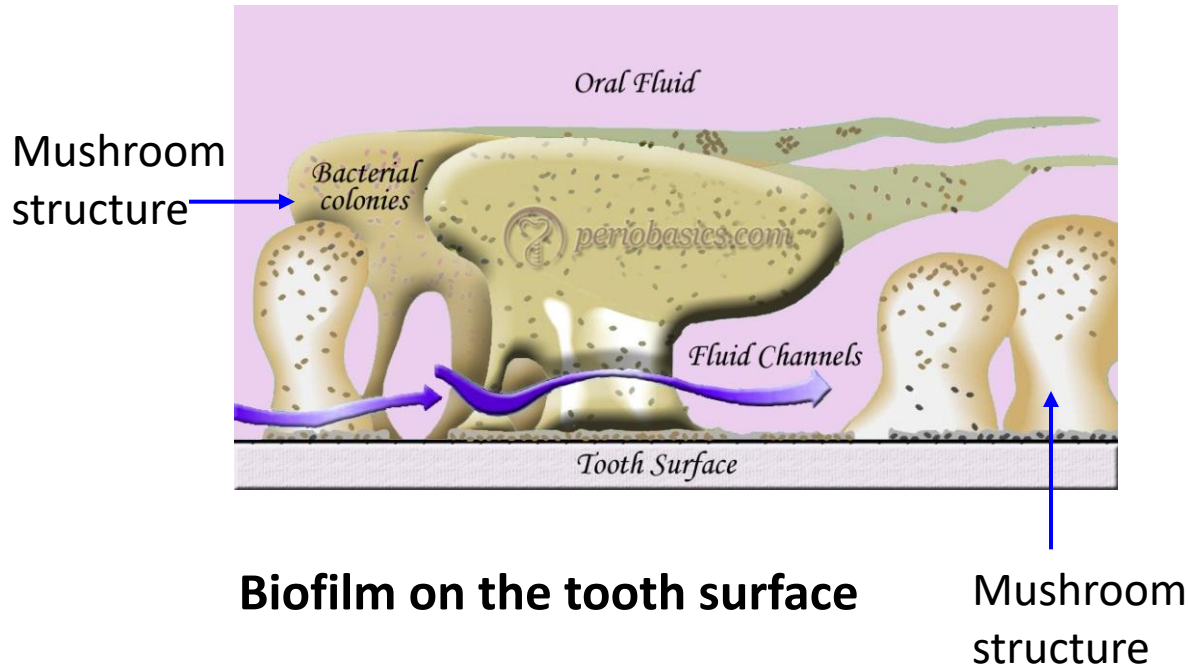
- enclosed in an extracellular polymeric substances 胞外多糖(**EPS**)
- adhere to an inert (non-living) or living surface
- structure is complex with “mushrooms” and channels carrying fluids



Microbes colonize surfaces through formation of biofilms

In medicine, biofilms cause serious infection and illness

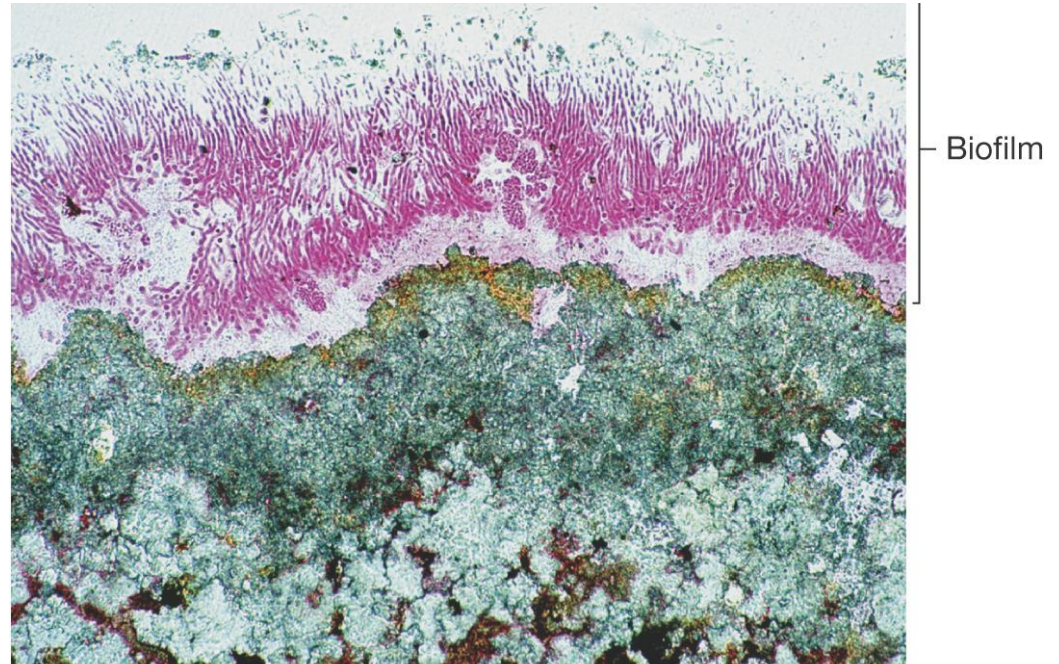
- Biofilms are found on our teeth (called plaque 菌斑)
- Biofilms are found on medical devices e.g. hip (髋关节) and knee implants (移植)



Microbes colonize surfaces through formation of biofilms

In the environment, most wet surfaces are covered with biofilm

- marine, fresh water, moist terrestrial (潮湿的地面)
- They are most often seen as layers of **slime** on
 - rocks
 - plants
 - animals
 - other microbes
 - man-made structures
e.g. hulls (外壳) of ships



PM Fig. 7.15

Biofilms form in a series of steps

1. Reversible

adhesion/attachment:

planktonic cells stick to a surface coated by proteins or other molecules from the environment

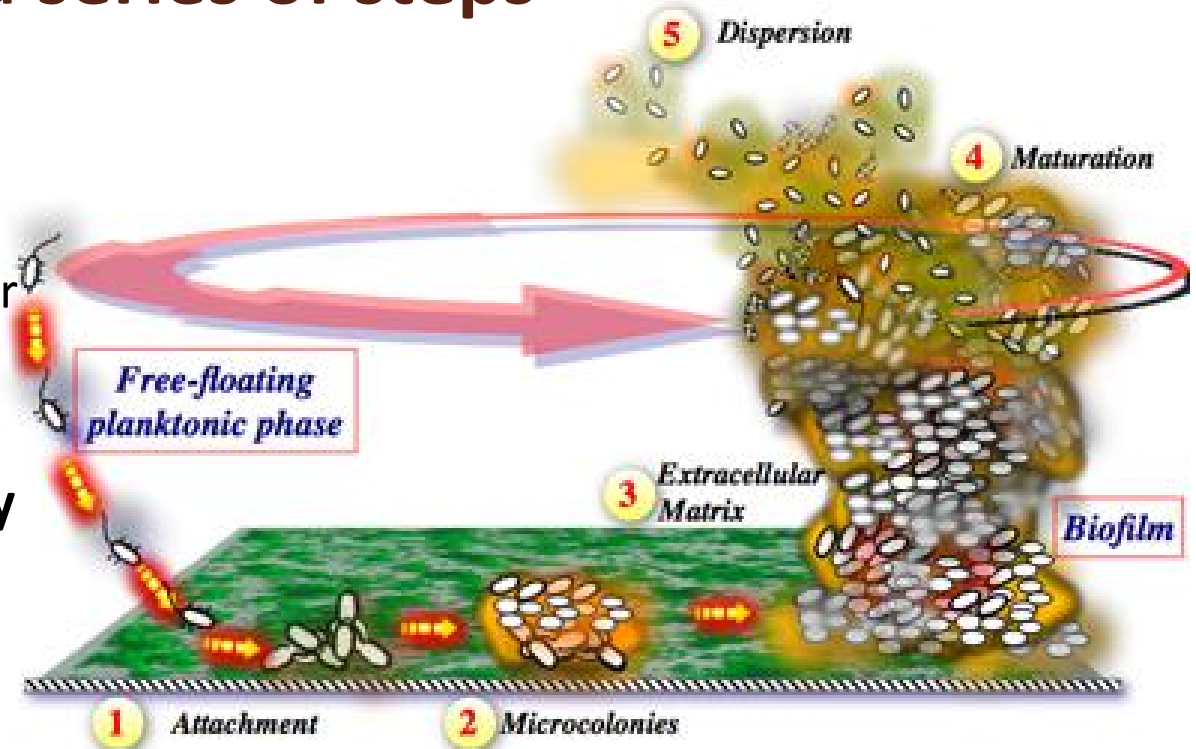
2. Colonization/microcolony

formation: the microbes stick more stably, form microcolonies

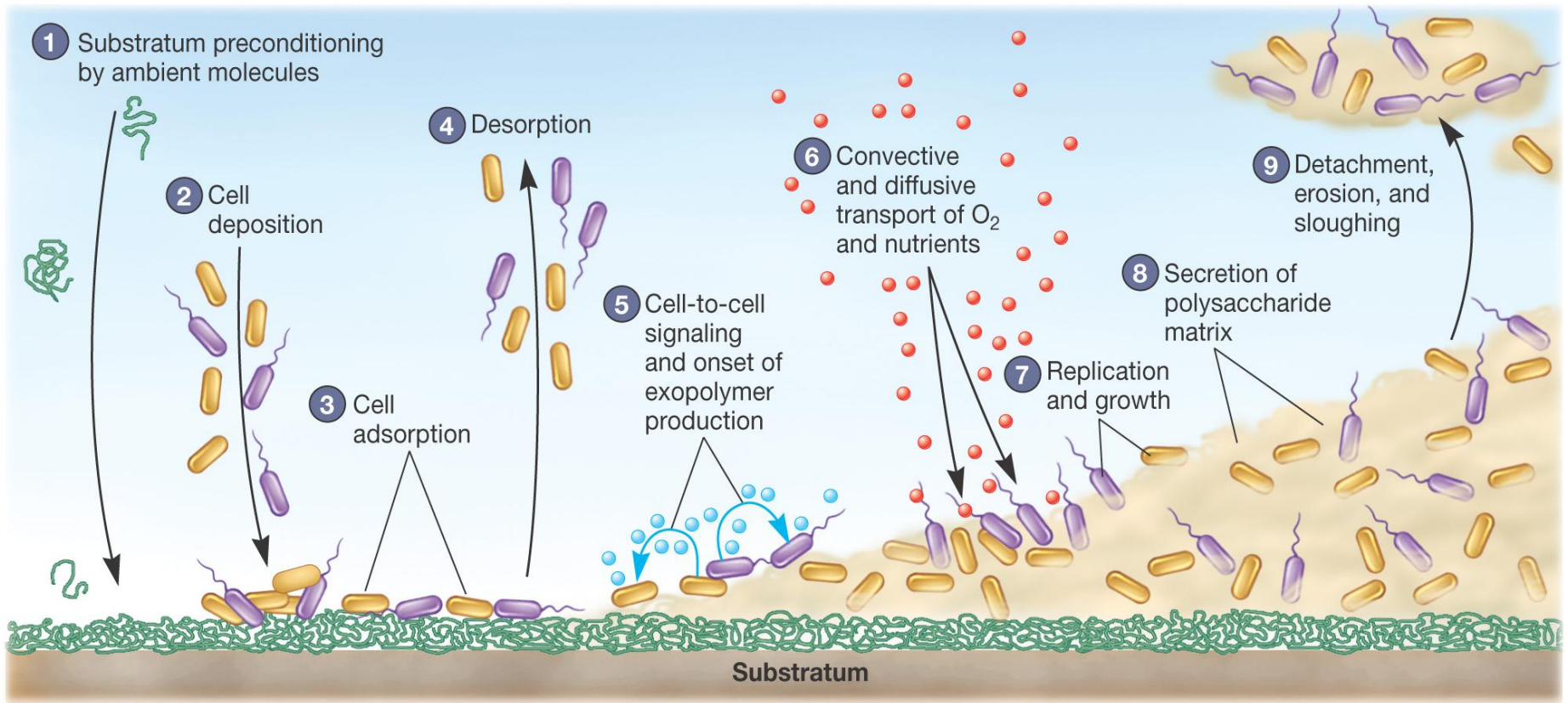
3. Extracellular matrix: the microbes grow and produce EPS e.g. polysaccharides, proteins. Other microbes may attach. (There is cell-to-cell communication, quorum sensing (群体感应), to be discussed in a later lecture)

4. Maturation (成熟): more EPS is produced. The biofilm thickens and “mushrooms” form.

5. Detachment/dispersion: the cells in a mature biofilm may become detached and return to a free-swimming, planktonic form



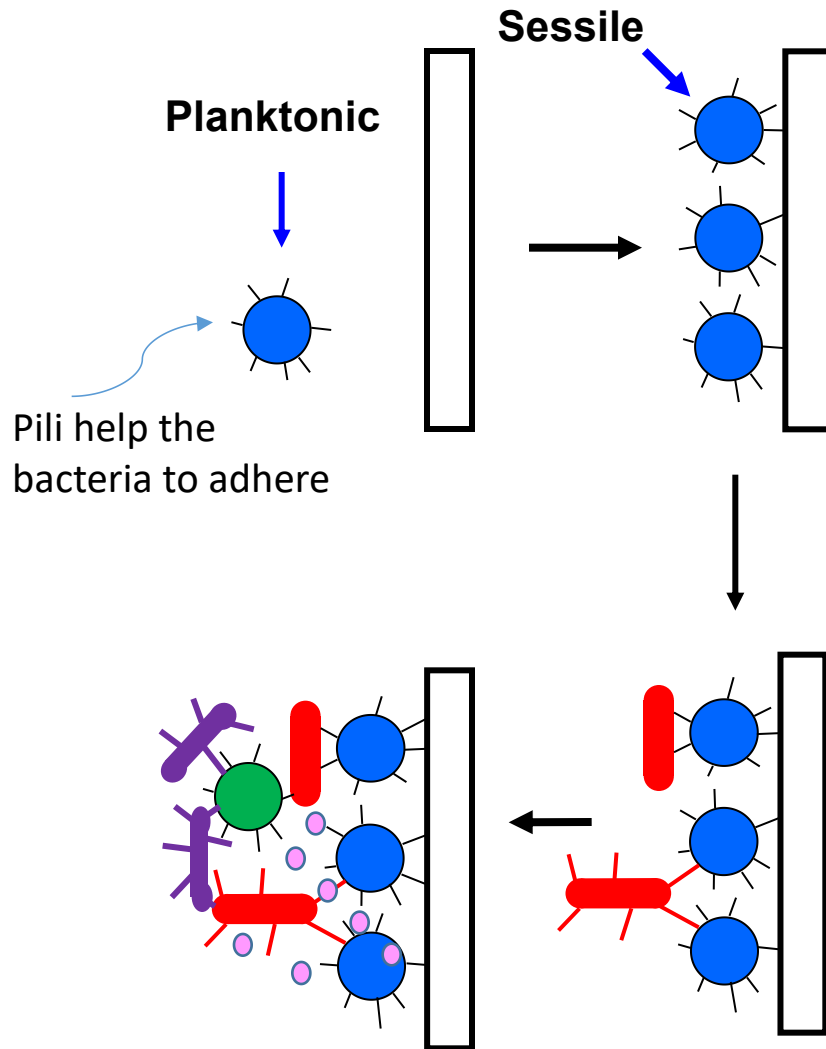
Biofilms form in a series of steps



PM Fig. 7.16

More steps are shown – no need to learn all these, learn the FIVE steps shown in slide 24. But this figure from Prescott's Microbiology gives a good idea of how a biofilm forms.

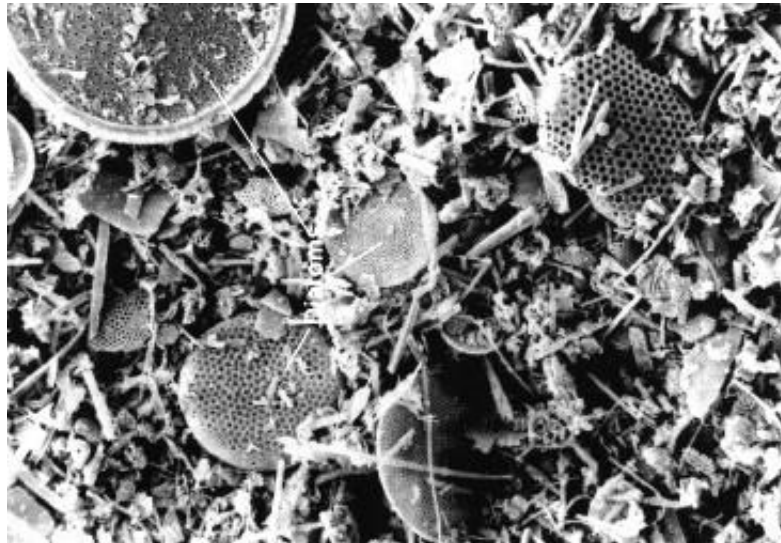
Microbes colonize surfaces through formation of biofilms



- In marine environment, biofilms are usually composed of many species
- Bacteria, algae, flagellates, ciliates, other protists, and viruses may all be found in a mature biofilm
- Advantages of biofilm for bacteria
 - avoid predation (捕食)/feeding by protists
 - genes are easily transferred from cell to cell
 - bacteria are protected from harmful agents e.g. UV

Microbes play a key role in the formation of sediments

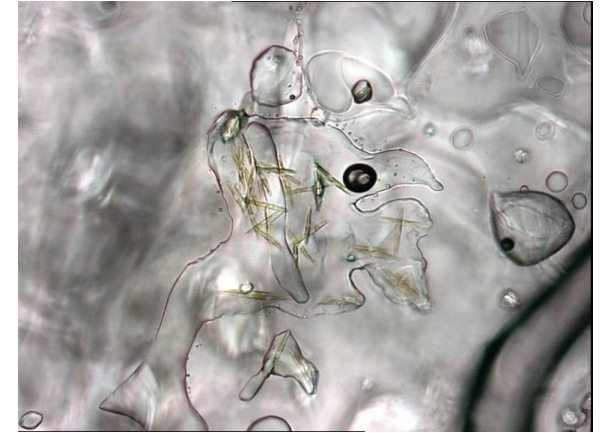
- In the deep ocean, about 75% of the ocean floor is covered by abyssal clays 深海黏土 (dust, volcanic ash) and oozes (软泥)
- Biogenous oozes contain >30% biological material
 - Calcium carbonate shells of protists – coccolithophorids (球石藻) and foraminifera (有孔虫)
 - Silica shells (frustules) of protists – diatoms and radiolarians



Other important marine environments where microbes are found

Sea ice

- 10% of Earth's surface, temp. as low as -20°C , brine (盐水) channels in the ice salinity 150 ‰, liquid at -35°C
- colonized by mixed communities of psychrophilic (嗜冷的) photosynthetic and heterotrophic protists and bacteria, and viruses
- dominant photosynthetic organisms near the ice-sea interface (交界) are pennate diatoms (羽衣硅藻) and small dinoflagellates (鞭毛藻)
- important part of the food chain e.g. food for krill 磷虾 (shrimp-like crustaceans 甲壳动物, main diet for fish, birds, mammals in polar oceans)



Algal chains and filaments in ice



Ice core showing algae band.
<http://oceanexplorer.noaa.gov/explorations/15arctic-microbes/welcome.html>

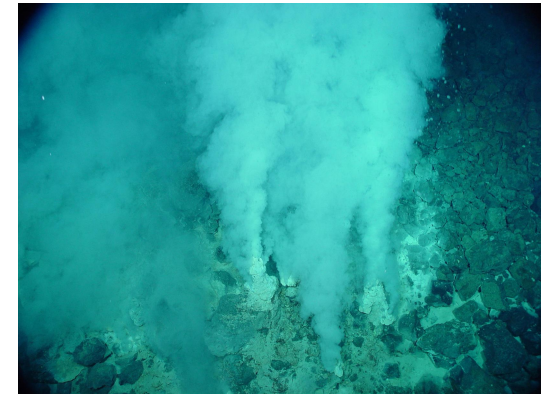
Other important marine environments where microbes are found

Hydrothermal vents (深海热泉)

- A fissure (crack 裂缝) in a planet's surface from which geothermally (地热) heated water issues.
- They are commonly found near volcanically active places, areas where tectonic plates (地壳板块) are moving apart. New earth crust being made.
- A **black smoker** is a type of hydrothermal vent found on the sea floor. They appear as black, chimney-like structures (up to 60 m) that emit a cloud of black material, with high levels of metal sulphide (硫化物) and sulphates (硫酸盐)
 - Temperatures up to 350°C
 - Anoxic



Black smokers



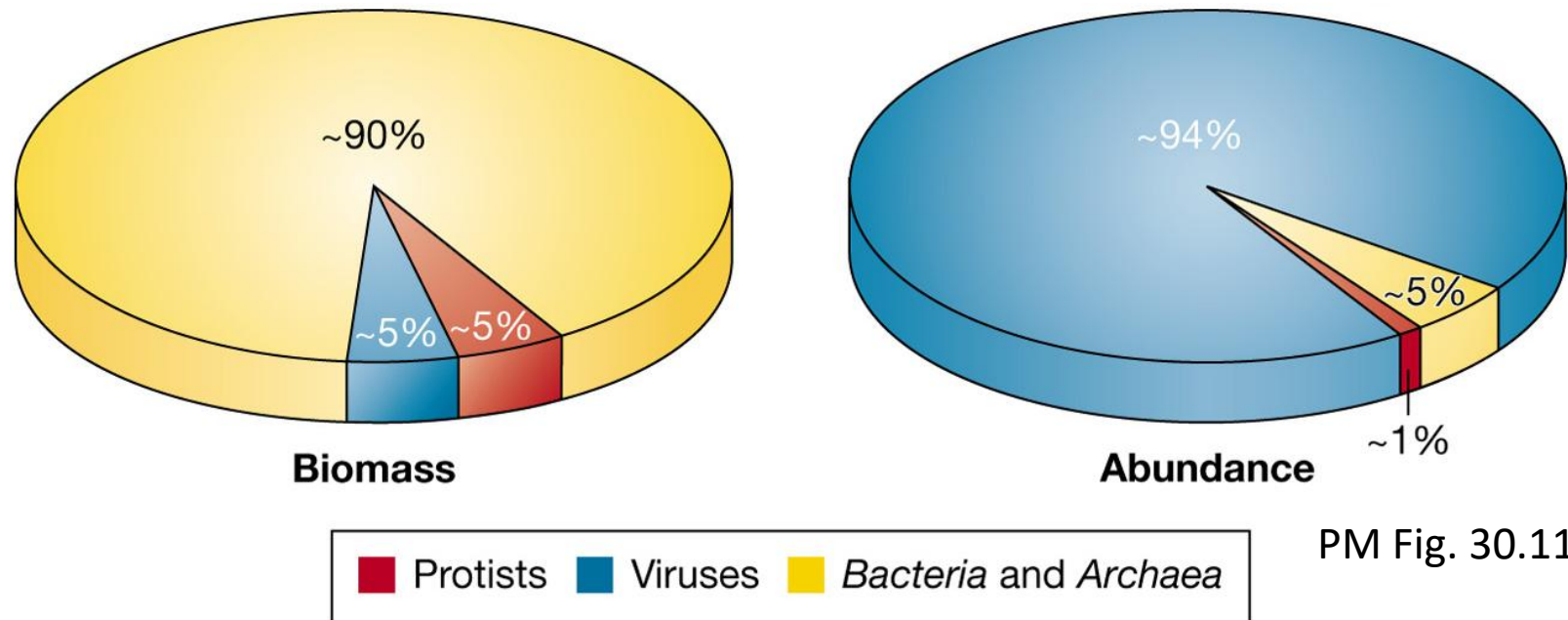
White smokers – lower temperatures , Ba, Si, Ca

Hydrothermal vents

- Discovery of life around the vents was totally unexpected!
 - tubeworms (管虫), clams (蛤蜊), anemones (海葵), crabs etc.
- Many of these animals contain chemosynthetic bacteria as **symbionts** (共生体) within their tissues, or on their surfaces (covered in Symbiosis lectures later)
- Also, warm waters near vents contain large populations of bacteria and archaea
 - they fix CO_2 using energy from the oxidation of sulphides
 - a temperature gradient exists around vents, so diverse communities of thermophiles and hyperthermophiles with different temperature optima occur
 - More Archaeal species than Bacterial species are hyperthermophiles: most have monolayer membrane which is more stable at high temps



Relative biomass and abundance of marine microbes



- a) Biomass of bacteria and archaea far exceeds that of both protists and viruses
- b) Counting individual organisms, the vast majority of marine microbes are viruses!

SUMMARY

- ♦ Marine microbes are found in all 3 domains of life, plus viruses
- ♦ Sizes are diverse, but extremely small cells are common in oceans
- ♦ Factors affecting microbes in the oceans include pH, C, light, temperature, O₂, salinity
- ♦ Marine microbes form a major part of plankton, which is classified by size
- ♦ Marine microbes form a major part of marine snow
- ♦ Many microbes in the environment live in structured communities called biofilms
- ♦ Other important marine environments for microbes are sediments, sea ice and hydrothermal vents